

# Mario Suarez Ayala

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Robotics • Embedded Systems • Electronics • Mechanics • Renders

## PROJECT DOCUMENTATION

|            |                                                                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project    | Project Vulcan                                                                                                                                                                                                |
| Version    | 1.0.0                                                                                                                                                                                                         |
| Date       | May 7, 2026                                                                                                                                                                                                   |
| Author     | Mario Suarez Ayala                                                                                                                                                                                            |
| Repository | <a href="https://github.com/MarioSuarezAyala1/Assistive-Electric-Wheelchair-System-Collaborative-Project">https://github.com/MarioSuarezAyala1/Assistive-Electric-Wheelchair-System-Collaborative-Project</a> |
| Type       | Embedded / Electronic / Mechanical                                                                                                                                                                            |
| Status     | Concept / Development Stage                                                                                                                                                                                   |



## Project Overview

Assistive Electric Wheelchair System is a collaborative engineering project focused on the development of an electric wheelchair platform for paraplegic users. The project integrates embedded electronics, mechanical design, control systems, and power distribution into a modular mobility solution.

I joined during the mid-stage of development, contributing mainly to the design of a Pelican-style enclosure for control and power electronics, internal component organization, 3D assemblies, renders, and a proposed electronic architecture for system integration.

The project was led and fully integrated by Engineer Dr. Urbina Norberto, who coordinated the overall architecture and final system implementation. Although an electronic architecture proposal was developed as part of my contributions, only a small portion of that proposal was implemented in the final version.

## Technical Specifications

Fill in the table with the key system parameters. Add or remove rows according to the project type.

| Parameter            | Value / Description             | Unit |
|----------------------|---------------------------------|------|
| Main Controller      | ESP32                           | —    |
| Communication        | Embedded / Wireless             | —    |
| Power System         | Battery-based                   | —    |
| Control Interfaces   | Physical emergency buttons      | —    |
| Electronic Enclosure | Pelican-style rigid case        | —    |
| Architecture         | Modular electronic distribution | —    |
| 3D Design Tool       | SolidWorks                      | —    |
| Render Engine        | Blender Cycles                  | —    |

## System Architecture

- **Mechanical Subsystem**

- Development of a rigid Pelican-style enclosure designed to protect and organize control and power electronics. Internal mechanical distribution and assembly concepts were created for modularity and maintenance accessibility.

- **Electronic Subsystem**

- Proposal for an embedded electronic architecture integrating ESP32 modules, emergency controls, sensors, display systems, and power distribution.

- **Render / Visualization**

- 3D assemblies and renders were developed in Blender and SolidWorks to visualize component integration and enclosure design before implementation.



## Bill of Materials (BOM)

| Component               | Specification              | Qty | Reference |
|-------------------------|----------------------------|-----|-----------|
| ESP32 Modules           | Embedded controller        | —   | —         |
| Emergency Buttons       | Physical control interface | —   | —         |
| Display System          | Monitoring interface       | —   | —         |
| Battery System          | Power source               | —   | —         |
| Pelican-style Enclosure | Electronics protection     | 1   | —         |



## Installation & Usage

- Fabricate the structure following CAD design
- Assemble and weld all structural components
- Install wheels and verify alignment
- Apply surface finishing and paint
- Load tools ensuring proper weight distribution



## Results & Testing

The project achieved a functional proposal for electronic integration and protective mechanical housing for embedded systems. The enclosure concepts and assemblies improved organization, modularity, and accessibility of critical electronics.



## Contributing

This project was developed collaboratively under the leadership of Engineer Dr. Urbina Norberto.

## My Contributions:

- Pelican-style enclosure design
- Internal electronic organization
- 3D assemblies and renders
- Partial electronic architecture proposal
- Physical control interface concepts



## License

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## Contact

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